

for any $\gamma \in \Gamma$. This last fact, along with compactness of Γ and S , continuity of $\pi(\cdot)$, and Lemma 2.4 in Newey and McFadden (1994) guarantees that $E[m_\gamma]$ is a continuous function of γ and that

$$\sup_{\gamma \in \Gamma} \left\| \frac{1}{n} \sum_{i=1}^n m_\gamma(Z_i) - E[m_\gamma(Z)] \right\| \xrightarrow{P} 0.$$

These conditions are sufficient for $\hat{\gamma} \xrightarrow{P} \gamma_0$ (see, e.g. Newey and McFadden (1994, Theorem 2.1)).

Now, let us study the asymptotic distribution of $\sqrt{n}(\hat{\gamma} - \gamma_0)$. Assumption 4.2 along with Lemma 7.6 in van der Vaart (1998) guarantees that, for $p_\gamma = \pi(X'\gamma)^D(1 - \pi(X'\gamma))^{(1-D)}$, the map $\gamma \mapsto p_\gamma^{1/2}$ is differentiable in quadratic mean with derivative

$$\dot{l}_\gamma = X \frac{\dot{\pi}(X'\gamma)}{\pi(X'\gamma)(1 - \pi(X'\gamma))} (D - \pi(X'\gamma)).$$

Therefore, the map $\gamma \mapsto p_\gamma^{1/2}$ is differentiable in quadratic mean at γ_0 with derivative $\dot{l}_{\gamma_0}$. Take a convex open neighbourhood N_{γ_0} of γ_0 contained in Γ . By Assumption 4.2(ii) and (iv), $\|\partial m_\gamma / \partial \gamma\|$ is bounded by a constant on N_{γ_0} . Therefore, by Theorem 9.19 in Rudin (1976), $|m_{\gamma_1} - m_{\gamma_2}| \leq M\|\gamma_1 - \gamma_2\|$ for any $\gamma_1, \gamma_2 \in N_{\gamma_0}$ and for some constant M . Finally notice that, by Assumption 4.2(ii)–(iv), $E[\dot{l}_{\gamma_0} \dot{l}_{\gamma_0}'] = E[\{\dot{\pi}_0^2 / (\pi_0(1 - \pi_0))\} X X']$ is non-singular. Therefore, by Theorem 5.39 in van der Vaart (1998), we obtain $n^{1/2}(\hat{\gamma} - \gamma_0) = n^{-1/2} \sum_{i=1}^n \psi_{\gamma_0}(Z_i) + o_p(1)$.

Let $m(Z, \beta, \gamma) = X_k \pi(X'\gamma)[\varphi(Z, \gamma)Y - X'_k \beta]$. By Assumption 4.2(vi), $E_M[m(Z, \beta, \gamma_0)]$ has a unique zero at $\beta_0 = (E[X_k \pi_0 X'_k])^{-1} E_M[X_k \pi_0 \varphi_0 Y]$. Under the assumptions of the theorem, the function $m(Z, \beta, \gamma)$ is continuous at each $(\beta, \gamma) \in \Theta \times \Gamma$. Since $\pi(X'\gamma)$ is bounded away from zero and one, and both Y and $\|X_k\|$ have finite first moments, then $\|m(Z, \beta, \gamma)\|$ is dominated by a variable with finite first moment. Therefore, by Lemma 2.3 in Newey and McFadden (1994)

$$\sup_{(\beta, \gamma) \in \Theta \times \Gamma} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \gamma) - E_M[m(Z, \beta, \gamma)] \right\| \xrightarrow{P} 0, \quad (\text{A.6})$$

and $E[m(Z, \beta, \gamma)]$ is continuous at each $(\beta, \gamma) \in \Theta \times \Gamma$. By the Triangle inequality,

$$\begin{aligned} \sup_{\beta \in \Theta} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \hat{\gamma}) - E[m(Z, \beta, \gamma_0)] \right\| &\leq \sup_{\beta \in \Theta} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \hat{\gamma}) - E[m(Z, \beta, \hat{\gamma})] \right\| \\ &\quad + \sup_{\beta \in \Theta} \|E[m(Z, \beta, \hat{\gamma})] - E[m(Z, \beta, \gamma_0)]\|. \end{aligned} \quad (\text{A.7})$$

Equation (A.6) implies that the first term of the R.H.S. of equation (A.7) is $o_p(1)$. Continuity of $E[m(Z, \beta, \gamma)]$ and consistency of $\hat{\gamma}$ give pointwise convergence for the second term of the R.H.S. of equation (A.7), uniform convergence holds by Θ being compact. These conditions guarantee $\hat{\beta} \xrightarrow{P} \beta_0$.

It can be shown that $\|m(Z, \beta_1, \gamma_1) - m(Z, \beta_2, \gamma_2)\| \leq \|X_k\|^2 \cdot \|\beta_1 - \beta_2\| + (C \cdot |Y| \cdot \|X_k\| + \|X_k\|^2 \|\beta_2\|) \cdot \|\pi(X'\gamma_1) - \pi(X'\gamma_2)\|$, for some $C > 0$ and $\beta_1, \beta_2 \in \Theta$. Using Theorem 9.19 in Rudin (1976), it is easy to show that for γ_1, γ_2 in some open ball containing γ_0 , and $X \in S$, $\|\pi(X'\gamma_1) - \pi(X'\gamma_2)\| \leq M\|\gamma_1 - \gamma_2\|$, for some $M > 0$. This Lipschitz property, along with the existence of finite second moments for Y and X_k , implies that the class of functions $\{m(Z, \beta, \gamma) : \|\beta - \beta_0\| < c, \|\gamma - \gamma_0\| < c\}$ is Donsker for some $c > 0$ (see, e.g. van der Vaart (1998, p. 271)). In addition, $E_M\|m(Z, \beta, \gamma) - m(Z, \beta_0, \gamma_0)\|^2 \rightarrow 0$ as $(\beta, \gamma) \rightarrow (\beta_0, \gamma_0)$. Existence of finite second moments for $\|X_k\|$ also implies that $\partial E_M[m(Z, \beta, \gamma)] / \partial \beta = -E[X_k \pi(X'\gamma) X'_k]$ which is non-singular in a neighbourhood of γ_0 by continuity. Notice that M_{γ_0} is the derivative of $E_M[m(Z, \beta_0, \gamma)]$ at γ_0 . Applying the delta method: $n^{1/2} E_M[m(Z, \beta_0, \hat{\gamma})] = n^{1/2} M_{\gamma_0}(\hat{\gamma} - \gamma_0) + o_p(1)$. Now, since $\hat{\gamma}$ is asymptotically linear, apply Theorem 5.31 in van der Vaart (1998) to get

$$n^{1/2}(\hat{\beta} - \beta_0) = Q^{-1} \frac{1}{n^{1/2}} \sum_{i=1}^n m(Z_i, \beta_0, \gamma_0) + M_{\gamma_0} \psi_{\gamma_0}(Z_i) + o_p(1).$$

Then, existence of second moments of ψ implies the result of the theorem. $\parallel$

Proof of Theorem 4.4. Like in the proof of Theorem 4.2, notice that

$$\begin{aligned} &X_{ki} \hat{\pi}(X_i) (\hat{\varphi}_i Y_i - X'_{ki} \hat{\beta}) - X_{ki} \pi_0(X_i) (\varphi_{0i} Y_i - X'_{ki} \beta_0) \\ &= X_{ki} \left(\frac{T_i - \lambda}{\lambda(1 - \lambda)} \cdot \frac{D_i - 1}{(1 - \hat{\pi}(X_i))(1 - \pi_0(X_i))} \right) Y_i (\hat{\pi}(X_i) - \pi_0(X_i)) - X_{ki} X'_{ki} (\hat{\beta} - \beta_0). \end{aligned}$$

By the Lipschitz property shown above for $\pi(\cdot)$, and $E_M Y^2 < \infty$, we obtain

$$\frac{1}{n} \sum_{i=1}^n \|m(Z_i, \hat{\beta}, \hat{\gamma}) - m(Z_i, \beta_0, \gamma_0)\|^2 = o_p(1). \quad (\text{A.8})$$

Continuity of $\hat{\pi}$ and Lemma 4.3 in Newey and McFadden (1994) imply that $\|\hat{M}_{\hat{\gamma}} - M_{\gamma_0}\| = o_p(1)$. Using similar arguments, it can be easily seen that boundedness of $\hat{\pi}$ implies that $n^{-1} \sum_{i=1}^n \|\hat{\psi}_{\hat{\gamma}}(Z_i) - \psi_{\gamma_0}\|^2 = o_p(1)$. Apply the Triangle and Hölder's inequalities to obtain the desired result. $\parallel$

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